Handoff to Researcher B

Dataset delivery, and how it compares with the project Executive Summary

Item	Value
From	Researcher A (data acquisition, cleaning, preprocessing)
To	Researcher B (GARCH-EVT, quantile regression, evaluation)
Date	2026-08-23
Delivered	Datasets/ — 6 indices, 96 columns, 1990-2026, fully documented
Status of the data	COMPLETE and validated: 1,195 + 158 checks, 0 failures
Status of the models	NONE FITTED. No model has been estimated.
Reference plan	Executive Summary.pdf, project root

1. Read this first — the handoff is not a clean transfer

The Executive Summary divides the work between two researchers, and it assigns **more to Researcher A than data**. Specifically, A leads:

Researcher A module	Budget	Due	Status
Baseline GARCH & GJR/EGARCH coding	24 h	Week 4	NOT STARTED
Realized GARCH model coding	24 h	Week 6	NOT STARTED
Rolling out-of-sample forecast engine	24 h	Week 8	NOT STARTED
Robustness checks	16 h	Week 12	NOT STARTED

That is 88 hours of Researcher A work still outstanding, and three of the five model-and-engine modules. Handing the whole remaining project to Researcher B would move roughly a third of the total budget onto one person and leave B leading every module, which is not what the plan agreed. Either re-agree the split with B in writing, or A retains the baseline GARCH, Realized GARCH and rolling-engine work. Flagging it here rather than letting it surface at the week-8 checkpoint.

What IS complete is the data half: acquisition, realized-volatility construction, cleaning, exploratory analysis and documentation — the two Researcher A deliverables due in weeks 2 and 3, both exceeding their specified scope.

2. Task status against the plan

Task area	Lead	Budget	Status	Notes
Data acquisition (daily & intraday) & cleaning	A	40 h	COMPLETE	Six indices, 1990-2026 daily; intraday 2011-09 onward (DAX 2013-09). 1,195 acquisition checks, 0 failures.
Realized volatility construction	A	16 h	COMPLETE	RV at five frequencies plus semivariance, MedRV, quarticity, subsampled RV, jumps. Exceeds the plan, which specified only RV = sum of squared intraday returns.
Baseline GARCH & GJR/EGARCH coding	A	24 h	NOT STARTED	Researcher A leads this in the plan. EDA has established the specification: Engle-Ng rejects on all six indices, so GJR or EGARCH is required, not plain GARCH.
Realized GARCH model coding	A	24 h	NOT STARTED	Researcher A leads. Inputs are ready; the measurement equation must absorb ScaleFactor_HL.
Rolling out-of-sample forecast engine	A	24 h	NOT STARTED	Researcher A leads. Must handle the Nikkei 2016-17 RV gap by using the last AVAILABLE observations rather than the last N calendar days.
Robustness checks (thresholds, windows, horizons)	A	16 h	NOT STARTED	Researcher A leads. Samples A and C are pre-built as ready-made robustness runs.
Environment setup & reproducibility	A	part of 16 h	PARTIAL	requirements.txt exists and all 26 scripts are numbered and rerunnable. NO git repository, no Dockerfile, no seed policy yet. The 'arch' package is not installed.
GARCH-EVT (filtering + EVT tail fitting)	B	24 h	READY TO START	Threshold diagnostics done: recommend the 95th-97.5th percentile, 230-460 exceedances. Must be re-estimated on genuine GARCH residuals.
Quantile regression (with predictors)	B	16 h	READY TO START	Predictors screened, collinearity resolved, non-stationary levels flagged and differenced forms supplied.
Evaluation metrics (RMSE, QLIKE, VaR/ES backtest)	B	24 h	READY TO START	Realized measure available as the volatility proxy; use RV_Scaled for like-for-like comparison against squared daily returns.
Crisis/regime analysis	B	16 h	READY TO START	CrisisLabel (10 named windows), VolRegime and VolRegime_ExAnte now shipped in the dataset.
Statistical tests (Diebold-Mariano etc.)	B	12 h	READY TO START	BalancedRV_B flag supplied for pooled comparison; per-index tests should use each index's own RV_Valid days.
Figures & tables preparation	A/B	16 h	PARTIAL	Ten EDA diagnostic figures complete and documented. Results figures - forecast vs realized, VaR exceedance, performance heatmaps - cannot exist until models are fitted.

3. Deliverables and milestones

Deliverable	Week	Lead	Status	Note
Clean data files & code	2	A	DELIVERED	Exceeds scope: 6 indices, not 1-3
Realized volatility series	3	A	DELIVERED	Exceeds scope: extended measures
Baseline GARCH code	4	A	OUTSTANDING	Researcher A
GARCH-EVT code	5	B	OUTSTANDING	Researcher B — inputs ready
Realized GARCH code	6	A	OUTSTANDING	Researcher A
Quantile regression code	6	B	OUTSTANDING	Researcher B — inputs ready
Rolling forecast engine	8	A	OUTSTANDING	Researcher A
Evaluation scripts	9	B	OUTSTANDING	Researcher B
Initial results & figures	10	A/B	OUTSTANDING	Blocked on models
Regime-specific analysis	11	B	OUTSTANDING	Labels now shipped
Robustness analysis	12	A	OUTSTANDING	Samples A and C pre-built
Statistical comparison tests	12	B	OUTSTANDING	Researcher B
Code repository & environment	13	A/B	PARTIAL	No git repo yet
Reproducibility audit report	14	A/B	OUTSTANDING	Needs seeds + env lock
Handoff package for writers	15	A/B	OUTSTANDING	Blocked on results

2 of 15 delivered. Both are Researcher A data deliverables and both exceed their specified scope. The remaining 13 depend on models that do not yet exist.

4. Where the delivered data departs from the plan

Every deviation below is deliberate and has a reason. Two of them change what the paper can claim and must be carried into the write-up.

Area	Plan said	Delivered	Why	What it means for B
Assets	1-3 indices (S&P; 500, NASDAQ-100, maybe a commodity or FX)	6 indices across 5 regions: SPX, NDX, UKX, DAX, NKY, HSI	Broader regional coverage was requested. Also makes a Model Confidence Set across a six-index panel possible, which a 1-3 asset study cannot support.	Six times the model fits. Budget compute accordingly.
Intraday source	NYSE TAQ via WRDS (academic licence), or commercial tick feeds	Dukascopy index CFD, bid side, free	The project operates under a strict free-data constraint. TAQ requires a WRDS subscription.	RV is built from a PROXY, not the exchange index. Aligned correlation against the exchange daily return is 0.97-0.99. This must be disclosed in the paper.
Intraday coverage	2010-2020, 10+ years including 2008 GFC and 2011 Euro crisis	2011-09 onward; DAX only from 2013-09	This is where free intraday history actually begins. Verified by probing earlier years and receiving empty responses.	CRITICAL: the realized-measure models CANNOT see the GFC, the dot-com bust, the Asian crisis or the Euro sovereign crisis. GARCH-EVT and quantile regression can, since they estimate on daily data back to 1990. The three models therefore see different crisis histories and the paper must say so.
Daily coverage	2010-2025	1990-2026	Free and available; more tail observations.	Longer estimation sample for the daily-only models.
Risk indicators	VIX as an exogenous variable	17 implied-volatility indices plus 22 macro / risk-factor series	Each index gets a regional volatility index where one exists; macro factors support the quantile-regression feature set.	Richer predictor set, already screened for stationarity and collinearity.
Outlier handling	"Winsorize or filter erroneous ticks"	Erroneous INTRADAY sessions removed; daily RETURNS NOT winsorized at all	Winsorizing returns would destroy the object of study. It shrinks the estimated GPD shape parameter and deletes the exceedances that identify the tail, producing VaR that looks well-calibrated only because the violations were removed. The extreme days were individually verified as real market events.	DELIBERATE DEVIATION FROM THE PLAN. Researcher B must not reintroduce winsorization at the modelling stage.

Area	Plan said	Delivered	Why	What it means for B
Realized variance	RV = sum of squared intraday returns	That, plus realized semivariance, MedRV, bipower variation, realized quarticity, tripower quarticity, subsampled RV, jumps, realized skew and kurtosis	Signed and jump-robust measures are close to expected practice in a modern realized-measure tail paper.	HAR-Q and signed-jump specifications are available without new data work.
Crisis labels	Define event periods by dates or volatility thresholds	Both: 10 named CrisisLabel windows, plus VolRegime and a no-look-ahead VolRegime_ExAnte	The plan left the choice open; both answer different questions.	Regime analysis can start immediately. Note VolRegime uses full-sample cut-points and is for ex-post reporting only.
Storage	CSV or Parquet	CSV throughout	Universally readable, no engine dependency.	Files are larger. Convert to Parquet if load time becomes an issue.
Version control	All code in a Git repo, branches for major features	NOT a git repository	Not yet initialised.	GAP. Should be fixed before the modelling work starts, or the model code will have no history either.
Environment	requirements.txt or environment.yml, plus a Dockerfile	requirements.txt only	Partial.	GAP. Also: the 'arch' package the plan names is not installed.
Estimation window	Initial estimation window e.g. first 2000 days	Recommend 1000 days for the realized-measure models	Sample B has about 3,100 valid RV days per index. A 2000-day window would leave barely 1,100 forecast days and only ~11 expected exceedances at the 1% level.	Researcher B's call, but state it explicitly. 1000 days is also the McNeil-Frey choice.

5. The two deviations that matter most

5.1 The realized-measure models cannot see the 2008 crisis

The plan called for 10+ years of intraday data covering the GFC. Free intraday history begins in 2011 (2013 for the DAX), so this was not achievable at zero cost. The consequence is structural, not cosmetic:

CrisisLabel	Start	End	Days_in_SampleB	In_SampleB
Asian_Crisis_LTCM	1997-07-02	1998-12-31	0	False
DotCom_Bust	2000-03-10	2002-10-09	0	False
GFC	2007-08-01	2009-06-30	0	False
Euro_Sovereign_Debt	2010-04-23	2012-07-26	0	False
China_Deval_Oil	2015-08-11	2016-02-11	128	True
Volmageddon	2018-02-02	2018-02-14	9	True
Q4_2018_Selloff	2018-10-01	2018-12-26	60	True
COVID_Crash	2020-02-20	2020-04-30	50	True
Rate_Shock_2022	2022-01-03	2022-10-12	196	True
Yen_Carry_Unwind	2024-08-01	2024-08-09	7	True

Sample B contains 6 of the 10 named crisis windows, about 14% of its days. It misses the Asian crisis, the dot-com bust, the GFC and the Euro sovereign crisis. GARCH-EVT and quantile regression still see all of them because they estimate on daily data from 1990; Realized GARCH does not. **State this asymmetry explicitly in the paper — a referee will otherwise read the comparison as unfair.** It does still contain COVID, the 2022 rate shock, the 2015-16 China devaluation, Volmageddon, Q4 2018 and the 2024 yen-carry unwind, which is enough distinct stress to identify a tail.

5.2 Returns are not winsorized, and must not be

The Executive Summary lists "Outliers: Winsorize or filter erroneous ticks" under preprocessing. Erroneous intraday sessions WERE removed. Daily returns were deliberately NOT winsorized, and this is a considered departure rather than an oversight. In a tail-risk study the extremes are the object of measurement: clipping them shrinks the estimated GPD shape parameter, deletes the exceedances that identify it, and produces VaR forecasts that appear well-calibrated precisely because the violations were removed. The largest returns were each verified as real market events. **Researcher B should not reintroduce winsorization at the modelling stage.**

6. What Researcher B receives

Item	What it is
01_ANALYSIS_READY/_analysis.csv	The model-ready dataset. 96 columns, 6 indices, 1990-2026, cleaned, gated and validated.
Dataset_Guide.pdf	14 pages. Section 3 is 24 precautions — read before coding.
Figure_Guide.pdf	Every diagnostic figure explained, with what it decides.
DATASET_MASTER_REPORT.xlsx	26 sheets: dictionary, precautions, all EDA tables.
EDA_REPORT.md	The full cleaning and exploratory analysis record.
CRISIS_PERIODS.csv	10 named crisis windows with justification for each date.
FEATURE_SETS.csv	Which fields feed which of the three models.
10_SCRIPTS/	26 numbered, rerunnable scripts covering the whole pipeline.

7. First eight steps

#	Step	Detail
1	Install the environment	pip install -r requirements.txt, then pip install arch. The plan names 'arch' for GARCH and VaR tests and it is currently absent.
2	Initialise version control	git init at the project root and commit the Datasets folder state before writing any model code, excluding 12_CACHE_REGENERATION (810 MB).

#	Step	Detail
3	Read Dataset_Guide.pdf section 3	Twenty-four precautions. The first four invalidate results outright if ignored.
4	Agree the sample and window	Sample B is primary: all six indices, 2013-09-30 onward, 2,685 common days. Recommend a 1000-day rolling estimation window.
5	Confirm the evaluation design	Per-index estimation on each index's own RV_Valid days; BalancedRV_B (1,994 days) only for pooled statistics. Do not force a balanced panel per-index.
6	Decide the non-synchronous-session convention	Asia closes before the US opens. Either adopt a lagged-information convention or estimate index by index and pool only losses. This must be settled BEFORE any cross-index result is produced.
7	Re-run the EVT threshold diagnostic	On genuine GARCH residuals. The shipped 95-97.5% recommendation comes from rolling-standardised returns as a stand-in.
8	Handle the Nikkei gap explicitly	No valid RV in 2016-17. Rolling windows must use the last AVAILABLE observations; no Realized-GARCH forecast can be produced or evaluated inside the gap.

8. Specification already settled by the EDA

These are not suggestions; they are conclusions from tests in the EDA report, and each is reproducible from the shipped validation tables.

Conclusion	Evidence
Plain GARCH(1,1) is insufficient	Engle-Ng sign-bias rejects on all six indices (worst p = 7e-5). Use GJR or EGARCH.
A Gaussian innovation is insufficient	Hill tail index 2.6-3.9, every estimate below 4 — the fourth moment does not exist.
The EVT stage is justified	GPD shape xi stays positive AFTER volatility standardisation, at 0.15-0.25.
Long memory is present	GPH d on log RV is 0.50-0.63; ADF and KPSS both reject.
An AR(1) mean term is warranted	Ljung-Box on returns rejects for 5 of 6 indices.
Use the level-plus-share realized block	LogRV with LogRS_neg gives VIF ~95; LogRV with RSV_Ratio, JumpShare and RSkew gives max VIF 8.1.
Differenced macro only	US10Y_pct and TermSpread_pct fail ADF in levels (p = 0.87, 0.33).
Implied vol leads for the tail	For the 1% quantile the vol index beats every realized measure; for the level of volatility the realized measures win.

9. Open questions for Researcher B

Question	Context
Rolling window length	1000 days recommended (McNeil-Frey). 2000 as in the plan would leave ~1,100 forecast days and ~11 exceedances at 1%.
Fixed or expanding window	The plan mentions both. Fixed is the cleaner comparison; expanding uses more of the 1990-2013 daily history.
Non-synchronous sessions	Must be settled before any cross-index result.
Half-days	Currently excluded from RV_Valid. Reversible via IsHalfDay if you prefer to keep them.
Pooled vs per-index MCS	BalancedRV_B supports pooling; per-index keeps more data.
ES backtest choice	The plan names Acerbi-Szekely. Not implemented.